

Dr. Carlos Manuel
HERRERA CASTILLO



CONTACT DETAILS

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Grimselstrasse, 1. Basel - CH

PERSONAL INFORMATION

Citizenship: **Spain**
Residency status: **Permit-C**
Family: **Single without children**
Languages: **Spanish** (Native), **French** (Family lang.), **English** (C2), **German** (B1)

RELEVANT SKILLS

- Genomics, bulk-RNAseq, scRNAseq, mutiomics.
- Sample fixation, DNA and RNA extraction, PCR, electrophoresis QC, library preparation.
- Use of bioinformatics packages to process and interpret genomic, transcriptomic and multiomic (scRNAseq + ATACseq) data.
- Ion chromatography, enzymatic kinetics, Northern, Southern and Western Blotting.
- Whole-mount skeletal staining.
- 2D and 3D X-ray, 3D output processing, landmark selection, procrustes analyses for morphometrics.

PROFILE

I am an evolutionary biologist specializing in the genomic, transcriptomic and developmental basis of morphological adaptation. My work focuses on connecting genotype to phenotype, identifying the regulatory mechanisms driving phenotypic divergence, combining large-scale genomic datasets, multi-omics approaches, and experimental genetics to have a complete understanding of the drivers of evolutionary change.

RESEARCH EXPERIENCE

PHD RESEARCH **06.2021 – 09.2025**
Department of Environmental Sciences (University of Basel).
♦ Objective: determining the genomic and transcriptomic causes of target phenotypes.
♦ Processed large-scale genomic datasets using high-performance computing clusters.
♦ Designed and executed complementation crosses to test candidate loci.
♦ Identified divergence stages via skeletal double-staining.
♦ Performed tissue sampling for bulk RNA-seq, single-cell RNA-seq, and single-cell ATAC-seq.
♦ Integrated multi-omics data to identify candidate regulatory elements and genes behind morphological trait loss.

MSC RESEARCH **09.2019 – 06.2021**
Paläontologisches Institut (University of Zurich).
Project 1: Domestication and Development in Birds:
♦ Investigated how domestication affects morphological development domestic birds.
Project 2: Description of New Fossil Material of *Laquintasaura venezuelae*:
♦ Described newly recovered fossil material from an early ornithischian dinosaur (*Laquintasaura venezuelae*).

EDUCATION

PHD IN ZOOLOGY **06.2021 – 09.2025**
University of Basel.
Specialization in genomics and transcriptomics.
MSC IN BIOLOGY **09.2019 – 06.2021**
University of Zurich.
BACHELORS IN BIOLOGY **09.2015 – 06.2019**
Universidad Autónoma de Madrid.

TEACHING EXPERIENCE

LECTURER: INTRODUCTION TO SCIENTIFIC WORK **06.2025 – Present**
University of Zurich.
CO-SUPERVISOR – MASTER’S THESIS PROJECT **02.2024 – 02.2025**
University of Basel.
LECTURER: INTRODUCTION TO BIOLOGY **09.2024 – 12.2024**
University of Basel.
INTRODUCTION TO LABORATORY PROTOCOLS **10.2023 – 04.2024**
University of Basel.

SOFTWARES

- R for genomic and transcriptomic processing
- STAR align (RNAseq aligner)
- NovoAlign and bwa-mem2 (Genomic aligners)
- AlphaFold
- Mimics (3D scan processing)
- HPC

AWARDS AND FUNDING

2025: Freiwillige Akademische Gesellschaft fellowship
2023 and 2024: The University of Basel travel fund for early career researchers
2019: Scholarship from Swiss-European Mobility Programme (SEMP)

CERTIFICATIONS AND LICENSES

LTK Module 20: Approved to perform animal experimentation on aquatic animals under Swiss law.

MEMBERSHIP OF PROFESSIONAL SOCIETIES

Member of the Society for the Study of Amphibians and Reptiles (SSAR)
Member of the Swiss Reproducibility Network
Reviewer in Journal *Molecular Ecology*

REFEREES

Dr. Daniel Berner (PhD supervisor): Universität Basel, daniel.berner@unibas.ch
Prof. Dr. Marcelo Sánchez-Villagra (MSc supervisor): Universität Zürich, m.sanchez@pim.uzh.ch
Prof. Dr. Dieter Ebert (co-supervised MSc student with): Universität Basel, dieter.ebert@unibas.ch

SELECTED PUBLICATIONS

Herrera-Castillo, Carlos Manuel, Brechbühl, Tanja, Fages, Antoine, Louise Dean, Laura, Tschopp, Patrick, MacColl, Andrew, Berner, Daniel **(2026) Evolution of threespine stickleback dorsal spines via *hoxdb* gene regulation.** DOI: 10.1101/2025.08.04.667941

Lampadaridis, Nikolaos, **Herrera-Castillo, Carlos Manuel*,** Ebert, Dieter* **(2026) Assessing planktivorous fish as vectors of a plankton parasite.** DOI: 10.64898/2026.07.09.737450
*Co-senior authors.

Herrera-Castillo, Carlos Manuel, Geiger, Madeleine, Núñez-León, Daniel, Nagashima, Hiroshi, Gebhardt-Henrich, Sabine, Toscano, Michael, Sanchez-Villagra, Marcelo R. **(2022) Skeletal variation in bird domestication: limb proportions and sternum in chicken, with comparisons to mallard ducks and Muscovy ducks.** 10.7717/peerj.13229

Herrera-Castillo, Carlos Manuel, Moratalla, José J., Belaústegui, Zain, Marugán-Lobón, Jesús, Martín-Abad, Hugo, Nebreda, Sergio M., López-Archilla, Ana I., Nebreda, Sergio M., Nebreda, Sergio M., Buscalioni, Ángela D. **(2022) A theropod trackway providing evidence of a pathological foot from the exceptional locality of Las Hoyas (upper Barremian, Serranía de Cuenca, Spain).** DOI: 10.1371/journal.pone.0264406

Herrera-Castillo, Carlos Manuel, Carrillo-Briceño, Jorge D. Sánchez-Villagra, Marcelo R. **(2021) Non-invasive imaging reveals new cranial element of the basal ornithischian dinosaur *Laquintasaura venezuelae*, Early Jurassic of Venezuela.** DOI: 10.5167/uzh-208139

CONFERENCES

Biology24

(University of Zürich - Switzerland)

2024

Oral presentation - Title: Investigating the causes of adaptation in three-spined sticklebacks.

3rd AsiaEvo Conference

(National University of Singapore - Singapore)

2023

Oral presentation - Title: Investigating the causes of adaptation in three-spined sticklebacks (a spiny subject)

10th International Conference on Stickleback Behaviour and Evolution
(University of Hólar - Iceland) 2022

Oral presentation - Title: Developmental genetic basis of skeletal evolution in stickleback fish.

Biology19

(University of Zürich - Switzerland)

2019

Poster presentation - Title: Skeletal variation in bird domestication.